

Mini E-Commerce Backend - Database Schemas

MongoDB + Mongoose Design Blueprint

1. User Schema

Field	Type
_id	ObjectId
name	String
email	String (unique)
password	String
role	String ("user" "admin")
timestamps	createdAt / updatedAt

Relationship: User -> Products (Reference), User -> Cart (Reference), User -> Orders (Reference)

Notes: Validation: name, email, password.

2. Product Schema

Field	Type
_id	ObjectId
title	String
description	String
price	Number
image	String
category	String
stock	Number
createdBy	ObjectId -> ref: "User"
timestamps	createdAt / updatedAt

Relationship: createdBy is a Reference to User.

Notes: Each product is created by an Admin user.

3. Cart Schema

Field	Type
_id	ObjectId
userId	ObjectId -> ref: "User"

products.productId	ObjectId -> ref: "Product"
products.quantity	Number (Embedded)
timestamps	createdAt / updatedAt

Relationship: userId and productId are References. quantity is Embedded.

Notes: Reference is used for Product because it already exists. Quantity belongs only to the cart.

4. Order Schema

Field	Type
_id	ObjectId
userId	ObjectId -> ref: "User"
products.productId	ObjectId -> ref: "Product"
products.quantity	Number (Embedded)
products.price	Number (Embedded)
totalPrice	Number
status	String
createdAt	Date

Relationship: userId and productId are References. quantity & price are Embedded.

Notes: Price is embedded to preserve the purchase price even if the product price changes later.

Reference vs Embedded Summary

Field	Design
Product.createdBy	Reference
Cart.userId	Reference
Cart.products.productId	Reference
Cart.products.quantity	Embedded
Order.userId	Reference
Order.products.productId	Reference
Order.products.quantity	Embedded
Order.products.price	Embedded